Teamwork Agreement Template

For the MathWorks Classroom Challenge Projects

Project: Battery Profile Charging

Team Name: BatteryTeam4

Instructor: Kevin Graham

Program/Class: EPP 2026 Cohort

Members and Roles:

1. Christie Ghazarian - Analysis/Validation Lead
2. Alexander Uribe - Modeling Lead
3. Vincenzo Cocciolone - Project Manager
4. Susana Zaragoza- Quality Assurance Lead
5. Split - Documentation & Visualization Lead

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations
- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs
- Documentation & Visualization Lead (SPLIT THIS): Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: Discord (e.g. Slack)

- Secondary channel: E-mail (e.g. E-mail for summaries/decisions)

- Response time: 24 on weekdays and 48 on weekends (e.g. within 24 hours on weekdays; within 48 hours on weekends)

- Meeting cadence: 60 mins on discord call on Sunday (e.g. weekly working meeting (60–90 minutes on Tuesdays via Zoom)

- Decision records: Google Spreadsheet (e.g. PM logs major decisions in a running document)

4) Collaboration & Tools

- Version control: GitHub and Google Drive (e.g. one shared GitHub repo or Google Drive folder)

- Data management: All datasets, parameters, and assumptions documented in

  README.md (e.g. README.md)

File naming & style: Snake_case (e.g. Snake_case, descriptive names)

- Backups: files sent to discord gc (e.g. PM tags old versions according to numbered system)

5) Quality Standards (Definition of 'Done')

Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report)

6) Timeline & Milestones

- Week 1: Team onboarding, problem framing, literature/background review, dataset/parameters

- Week 2: First pass models (hand calculations + baseline MATLAB scripts); initial plots/visuals.

- Week 3: Validation and quality assurance (support the team member with this role)

- Week 4: Report, slides, presentation rehearsal, repository cleanup (split this work evenly)

(e.g.
Week 1: Problem framing, literature/background review, dataset/parameters,
Week 2: First pass models (hand calculations + baseline MATLAB scripts); initial plots/visuals.
Week 3: Validation and quality assurance
Week 4: Report, slides, presentation rehearsal, repository cleanup)

7) Decision-Making & Conflict Resolution

- Default: discussion on call; consensus after discussion; PM facilitates (e.g. consensus after discussion; PM facilitates)

- If blocked >48 hrs: majority vote, PM documents decision (e.g. Time-limited debate, majority vote, PM documents decision)

- Technical disagreements: Evidence-based comparison (show that the code runs or logic works) before vote (e.g. Evidence-based comparison before vote)

- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

- Task board: To-Do / Doing / Review / Done with owners and due dates documented in README.md (this will updated every other day)

- Accountability: Missed deadline → recovery plan within 24 hrs; repeated misses trigger role redistribution.

- Attendance: If you'll miss a meeting, give 24-hour notice and share updates asynchronously (we will keep track of meetings on a google doc)

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- A fitted charging-curve model for a real lithium-ion battery dataset, including a graph of the data with the fitted curve and goodness-of-fit statistics
- Plots for: voltage vs. time, current vs. time, and power vs. time
- Analytical results (numerical values or derivatives)

- presented in a summary table Numerical integration results showing total energy delivered during charging.
- Time required to reach 80% and 100% charge.
- Rate-of-change analysis (derivatives at key intervals).
- Estimates of resistive energy loss.

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(e.g. Code repository with clear structure and README, hand calculations linked to modeled results for cross-validation, specific figures & visualizations, final report with specified sections)

12) Agreement & Signatures

Name & Signature: Christie Ghazarian

Date: 07/12/2026

Name & Signature: Susana Zaragoza

Date: 07/12/2026

Name & Signature: Alexander Uribe

Date: 07/12/2026

Name & Signature: Vincenzo Cocciolone

Date: 07/12/2026